P525/1

CHEMISTRY

Paper 1

2026

2¾ hours

Uganda Advanced Certificate of Education

CHEMISTRY

(Theory)

Paper 1

2 hours 45 minutes

INSTRUCTIONS TO CANDIDATES

*This examination paper consists of two sections; **A** and **B**. It has **six** examination items.*

*Section **A** has **two compulsory** items.*

*Section **B** has two parts; **I** and **II**. Respond to only **one** item from each part.*

*Respond to **four** items **in all**.*

*Any additional item(s) responded to will **not** be scored.*

*All responses **must** be written in the answer booklet(s) provided.*

Begin each item on a fresh page.

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SECTION A

Respond to both items from this section.

1. A small-scale gold-mining company operating near a river uses mercury to extract gold from crushed ore, and treats the resulting mercury-contaminated waste water with sodium sulphide before discharging it into the river, according to the equation:

$$Hg(NO_3)_2(aq) + Na_2S(aq) \rightarrow HgS(s) + 2NaNO_3(aq)$$

Grace, an environmental officer, received reports of dead fish along a stretch of the river close to the mining site. While investigating the mine's waste-treatment records, she found that the sodium sulphide is added in a fixed daily dose, regardless of how much ore is processed that day. Her findings for five consecutive days are shown in Table 1.

Table 1

Day	Mass of ore processed (kg)	Mass of $Hg(NO_3)_2$ formed in waste water (kg)	Mass of Na_2S added (kg)
Monday	500	3.24	2.40
Tuesday	500	3.24	2.40
Wednesday	1,600	10.37	2.40
Thursday	700	4.54	2.40
Friday	1,800	11.66	2.40

[Molar masses: $Hg(NO_3)_2$ = 324 g mol^{-1}; Na_2S = 78 g mol^{-1}; HgS = 232 g mol^{-1}]

The fish deaths were reported mainly on Wednesday and Friday evenings. Grace also noted complaints, on some other days, of a strong smell of rotten eggs around the discharge point.

Task:

a) Analyse Grace's findings, using mole calculations, to explain the reported fish deaths.

b) Assess the adequacy of the waste-treatment method used by the mining company and suggest how it could be improved.

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2. A soft-drinks bottling company preserves and carbonates its beverages using citric acid, sodium citrate and dissolved carbon dioxide gas. Retailers supplied by the company have raised two concerns: some bottles lose their fizz unusually quickly once opened, especially during hot weather, while a few other batches develop a mouldy smell on the shelf well before the stated expiry date.

The company's quality control chemist explained that the drink is prepared using either of two standard recipes, shown in Table 2, to make 500 cm^3 of drink.

Table 2

Substance	Recipe P	Recipe Q
Citric acid (g)	1.92	1.92
Sodium citrate (g)	0.294	0.294
Sugar (g)	60.0	15.0

The pK_a of citric acid is 3.13, and the recommended pH range for preserving and carbonating the drink is 2.8 – 3.2.

The dissolving of carbon dioxide gas in the drink during bottling is represented by:

$$CO_2(g) \rightleftharpoons CO_2(aq) \qquad \Delta H = -20 \text{ kJ mol}^{-1}$$

Bottles made with Recipe P are always chilled to about 4 °C soon after bottling and kept refrigerated, while several retailers who stock Recipe Q, especially those without reliable refrigeration, store their bottles at room temperature (about 28 °C) for long periods before sale.

Task:

a) Assess the retailers' concerns, using the data provided.

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b) Evaluate the measures the company could take to minimise the concerns raised, with reference to the principles of chemical equilibrium.

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SECTION B

Part I

Respond to any one item in this part.

3. A group of Senior Five students visited a lime-processing factory that supplies calcium oxide to a nearby water-treatment works and to a cement company. The tour guide explained that although both calcium carbonate and magnesium carbonate can be obtained cheaply from local quarries, only calcium carbonate is roasted to produce quicklime at the plant's usual furnace temperature; magnesium carbonate is not used this way. He added that strontium and barium carbonates, although available from a supplier, are avoided altogether at the plant.

He also mentioned that a small amount of beryllium oxide is used separately, to line special high-temperature furnace crucibles in one section of the factory. Unlike the other Group 2 oxides used on site, beryllium oxide does not dissolve to give an alkaline solution in the calcium-treatment tanks, but instead reacts with both dilute hydrochloric acid and dilute sodium hydroxide.

Table 3

Carbonate	Decomposition temperature (°C)	Hydroxide	Solubility in water at 25 °C (g per 100 g water)
$MgCO_3$	350	$Mg(OH)_2$	0.0009
$CaCO_3$	900	$Ca(OH)_2$	0.153
$SrCO_3$	1290	$Sr(OH)_2$	0.41
$BaCO_3$	1360	$Ba(OH)_2$	3.9

Having previously studied only Period 3 trends, the students found it difficult to explain why the furnace, set at a fixed temperature suited to calcium carbonate, would not be suitable for magnesium carbonate, and why beryllium oxide behaved so differently from the other Group 2 oxides mentioned.

Task:

a) Analyse the trends in properties of Group 2 elements and their compounds to explain the tour guide's observations.

b) Justify the unusual behaviour of beryllium oxide compared to the other Group 2 compounds mentioned, using the concept of diagonal relationships.

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4. An industrial chemistry class toured two factories in the same industrial area: a municipal water-treatment plant and a sulphuric acid manufacturing plant.

At the water-treatment plant, the manager explained that although fluorine, bromine and iodine can, in principle, also kill micro-organisms in water, the plant uses chlorine gas exclusively for disinfection. She provided the data in Table 4.

Table 4

Halogen	**$E°(X_2/X^-)$ / V**	**Bond energy of X–X / kJ mol^{-1}**	**Solubility in water (g per 100 g, 25 °C)**
F_2	+2.87	158	reacts violently with water
Cl_2	+1.36	242	0.73
Br_2	+1.07	193	3.41
I_2	+0.54	151	0.03

At the sulphuric acid plant, the guide described the Contact process, in which sulphur dioxide is oxidised to sulphur trioxide using vanadium(V) oxide as catalyst. He explained that the vanadium changes both its colour and its oxidation state during the catalytic cycle, alternating between V_2O_5 (yellow, vanadium in the +5 state) and VO_2 (blue, vanadium in the +4 state) as it first reacts with sulphur dioxide and is then reoxidised by oxygen.

The students found it difficult to explain why chlorine, rather than fluorine or iodine, was the practical choice for water treatment, and why a transition metal compound, rather than a Group 2 or Group 17 compound, was suitable as the industrial catalyst.

Task:

a) Analyse the trends in properties of Group 17 elements to justify the water-treatment plant's choice of chlorine for disinfection.

b) Justify the suitability of vanadium(V) oxide, rather than a Group 2 or Group 17 compound, as a catalyst in the Contact process, with reference to the properties of d-block transition elements.

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Part II

Respond to any one item in this part.

5. Joan, an environmental researcher, is studying plastic waste that collects along the shores of Lake Victoria and blocks fish-breeding grounds in shallow bays. She found that two types of plastic, R and Y, made up most of the waste and had clearly persisted in the environment for a long time without breaking down.

R is represented as $[–CH_2–CHCl–]_n$

Y is represented as $[–CH_2–CH(CH_3)–]_n$

On visiting a plastics factory near the lake, Joan learned that:

i. R is manufactured from chloroethene, which the factory produces by passing hydrogen chloride gas into ethyne in the presence of a mercury(II) chloride catalyst.

ii. Y is manufactured from propene, which the factory obtains by catalytic cracking of longer-chain alkanes supplied from a nearby oil refinery.

iii. Some residents living near the shore openly burn collected plastic waste, including R, and Joan noted that this produces a sharp, choking gas together with black smoke.

Joan requires detailed information about the mechanism of formation of chloroethene from ethyne, the polymerisation mechanism by which plastics R and Y are formed from their monomers, the cracking process that produces propene, and the environmental and health impact of the plastics and of burning them in the open.

Task:

a) Analyse Joan's observations about the plastic wastes collected from the lake shore.

b) Analyse the reactions involved in the synthesis of plastics R and Y and the other processes referred to in the scenario.

c) Evaluate the impact of plastics R and Y, and of burning them, on the environment and human health.

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6. A cosmetics and fragrance company in Uganda blends natural plant extracts with synthetic organic compounds to manufacture perfumes and skin-care products. The production manager explained that the company manufactures bromobenzene and phenylethanone from benzene and compound Y respectively, and manufactures propanone and isopropyl benzoate from compound X and benzoic acid respectively.

During a routine stock audit, the following compounds were found in the raw-materials store, each unlabelled apart from a data card:

Compound X: an alcohol with 3 carbon atoms, molecular mass 60, which turns acidified potassium dichromate(VI) from orange to green on gentle warming, and does not react further to form a carboxylic acid under the same conditions.

Compound Y: a chlorinated organic compound of molecular mass 78.5, which reacts vigorously with water to give a strongly acidic, fuming solution, and which reacts with benzene in the presence of anhydrous aluminium chloride to give a sweet-smelling product.

The store also contained a fourth, unlabelled liquid hydrocarbon which the store-keeper suspected might be either cyclohexene or benzene; on shaking a sample of it with bromine water in the absence of a catalyst, no decolourisation was observed even after several minutes.

The manager has asked for the raw materials to be positively identified, for a full account of how each of the products is manufactured, and for an evaluation of how safely and usefully these compounds can be handled in the factory.

Task:

a) Examine the compounds available in the store to determine their identities and their suitability as raw materials for the products described.

b) Analyse the reaction mechanisms and processes involved in the formation of bromobenzene, phenylethanone, propanone and isopropyl benzoate.

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c) Evaluate the practical applications and safety considerations of the various organic compounds kept in the store.

END

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